

# PERFORMANCE OF RFID TAGS IN NEAR AND FAR FIELD

Jose Luis Martinez Flores, Satya Sai Srikant, Bimal Sareen, Abhijit Vagga

AVAANA, New Delhi, India

## ABSTRACT

We observe some of the interrelationships that exist with respect to the performance of RFID Tags. We observe that in a classic system of wireless communication, the possible variables for the calculation of link performance are interrelated by diverse metrics. Once the link is established, it is important to characterize and register the performance of the system. It is this system's dependency on these diverse set of interrelated metrics, and their equations, which affects its performance. In essence, the performance is observed as a consequence of attenuation-optimized transmission and reception. We discuss the foundation of a near field and far field relationship model that helps confirm the interdependency of various factors as it relates to RFID Tag performance.

The frequency operated in Near field (125 KHZ, 13.56 MHz) is more reliable in many applications where Read Distance is not the major factor, and the properties of materials present near the surrounding of RFID system are affected less by the degradation of RF signal as Near Field works on the principle of Magnetic Coupling. In contrast, for far field, radiated signal gradually decreases the read distance depending upon the

condition of the environment and properties of materials / metals.

In this paper, we have concentrated on a simple way of measuring the Tag performance for RFID Technology considering all the parameters. The way presented here, it is based on RFID on implementation experience and Laboratory measurements.

## I. INTRODUCTION

To understand *RFID* technology characteristics one must first understand just what is RFID. RFID (Radio Frequency Identification) is a means of storing, and retrieving data through electromagnetic transmission to an RF compatible integrated circuit. Figure (1) shows the simple block diagram of RFID system. The RFID integrated circuit (reader) consists of a memory chip, but could also be designed where several electronic components together are used to form an integrated circuit design. The circuit includes one or several memory chips that are used for data storage, to substrate backing material or circuit board structure. Control logic circuits which may contain programmable and non-programmable integrated circuits to control the signals, integrated circuits to

modulate and demodulate, and an antenna tuned at a particular frequency are present. The tags contain the memory chips (acts as a Capacitance) along with the antenna (act as inductance). This antenna may contain different patterns / design for an operationally tuned frequency ( $f_o$ ) according to the shape and size of tags in order to have definite inductance value ( $L$ ) so that it must be tuned (to resonate) with the total capacitance value ( $C$ ) obtained by memory chip and parasitic capacitance occurred due to Controlled signal. It is given by Equation (1)

$$f_o = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

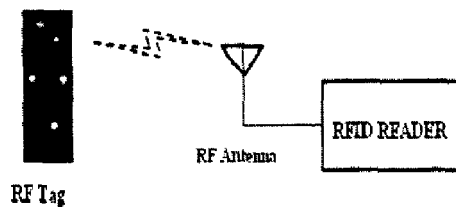


Figure 1. A simplified RFID system

RFID basically operates on two basic communication principles. The principles are: Wireless Communication (commercial Radio broadcasting, commercial TV broadcasting, WiFi, Mobile Communications, etc.) with Complete Modulation Techniques and secondly is the Microwave Link communication. Both of the techniques are linked among the systems containing the reader, antenna and the small tag. To operate the principle successfully, the cover of antennas must be sufficient to link the entire system with a Radiated Signal with undefined Signal Strength. The word undefined used is because while signal Radiates, it not only depend upon the above principles but also it depends upon the Environmental conditions as well as materials present closer to all the system. This Radiated

Signal is also required to energize the passive device (IC) present inside the Passive Tags. This signal must have enough signal strength to wake up the Passive tag and starts the proper communication of transferring the data (bit transfer) among the system otherwise with less signal strength, it communicates but will provide the distorted data or error data or missing data (Error data transfer). No such energize signal is required for Battery assisted Passive Tag (Semi Active) and Active tag as they already contain battery (source) to provide the sufficient energy for proper data transfer. In other words one could say that Passive tags lack circuit intelligence and Battery operated tag has sufficient intelligence for data transfer among the RFID system

## II. TYPES OF RFID TAGS

RFID tags can be differentiated into two major categories: *Passive* and *Active*.

The passive tag does not contain any internal power source such as a battery. This passive tag or "reflective" tag works on the principle based upon the electromagnetic energy radiated by an interrogator to power the RF integrated circuit that energizes the tag itself. Such types of tags are said to be "Beam Powered" and also known as "Batteryless -Contactless Tag". Due to this reason, such tags are very easier to design, easily available and less expensive to manufacture.

There is also a version of a passive tag that does contain a battery. This type of passive tag has some of the enhanced, and speed attributes of a true active tag, but still communicates in the same method, as do other passive tags. These active / passive tags that do contain an internal power source, usually are much more

complex integrated circuits with multiple components. Consequently, they are more expensive to make and purchase.

Finally, there are true active tags. Active tags contain both a radio transceiver and battery to power the transceiver. Because there is an inboard radio on the tag, active tags get substantially more range than passive or active / passive tags. Active tags are more expensive than passive tags and, as with any battery powered product, the batteries must be replaced periodically.

### III. RFID TAG PERFORMANCE

There are only two possible physics concepts used by RFID Technology for the detection of RF Tags. They are: Near Field Concept (Magnetic Coupling) and Far Field Concept.

The Near field systems employs inductive coupling of the transponder tag to the reactive energy circulating around the reader antenna where as the Far Field systems couple to the real power contained in free space propagating by electromagnetic plane waves. Near field coupling techniques are generally applied to RFID Systems operating in the LF and HF bands with relatively short reading distances well within the radian sphere defined by  $\lambda/2\pi$ , while Radiative Far-field coupling is applicable to potentially longer read range UHF and microwave RFID Systems. The wavelength of the frequency range operating at Near Field used (< 125 kHz: 2400 m, 13.56 MHz: 22.1 m) is several times greater than the distance between the reader's antenna and the transponder, the electro-magnetic field may be treated as a simple magnetic alternating field with regard to the distance between transponder and antenna. Hence the Near-field communication is based on

a magnetic field effect, which has an inverse sixth power ( $1/r^6$ ) relationship with range. For Far Field Communication, the electromagnetic radiation field effect has a ( $1/r^2$ ) relationship with the transmit and return path, i.e the read range of reflected signals from the tag to the reader. With reference to Figure (2), the reader emits a Radio Frequency (RF) wave by creating an electromagnetic field in the air. Near Field occurs at a distance less than  $\lambda/2p$ , where the magnetic field dominates. The Far Field occurs at a distance greater than  $\lambda/2p$ , where the electric field dominates. Here  $p$  is the boundary region between the Near Field and the Far Field. The Near Field RFID tag antenna is basically a coil; its physical size has little dependence on the RF used. The Far Field RFID tag antenna is usually a dipole that is half a wavelength long and is thus dependent on the frequency used. In general, the higher the frequency, the smaller the Far field RFID tag antenna.

*It is to be noted that the Near Field RFID such as RFID operates at 125 KHz and 13.56 MHz frequency is not suitable for the Far field RFID because it would need a tag antenna that is at least 10 m long which is impractical.*

The general RFID tags available at various frequencies are 125 KHz, 13.56 MHz, 433 MHz, 915MHz, 2450 MHz.. Table (1) shows the properties of RFID Tags at various frequencies. Depending upon the various orientations of antennas (tag), types of antennas based on the operating frequencies (LF / HF / VLF/ UHF), shape and size, mechanical integrity, Q, Bandwidth, Signal to Noise Ratio (SNR) proposed materials used, flat impedance matching of the system, the climatic condition, the noise present near the surroundings, presence of any metal near the field, the performance of the RFID

system matters. Moreover, metal and the liquid like water or blood are the biggest enemies of RFID system. This is the one of the biggest reason why passive RFID system are not usually advisable for Employee tracking at UHF and microwave frequencies.

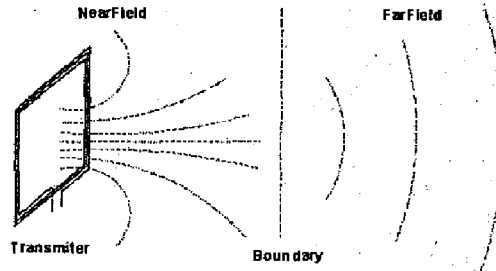


Figure 2. Field Pattern of RFID system

Table 1: RFID Properties

| Properties                          | LF<br>(125 KHz)   | HF<br>(13.56 MHz)       | UHF<br>(915/2450 MHz) |
|-------------------------------------|-------------------|-------------------------|-----------------------|
| Read Range                          | Low<br>(<24 Inch) | Moderate<br>(< 1.2 Mts) | High<br>(< 5Mts)      |
| Data Transfer Rate                  | Slow              | Moderate                | Fast                  |
| Read-Write data carriers            | Yes               | Yes                     | Yes                   |
| Directional Readers                 | No                | No                      | Yes                   |
| Read thru glass, clothes, wood etc. | Yes               | Yes                     | Yes                   |
| Dirt Resistance                     | Good              | Good                    | Good                  |
| Power Consumption                   | Low               | Moderate                | High                  |

The performance of RFID tags also depends upon Loaded Quality Factor ( $Q$ ) of antenna itself of the microchip which is the parallel combination of the loading effect on the tag antenna circuit by the power dissipation in the chip circuit ( $Q_{CHIP}$ ) and the quality factor of the antenna circuit only loaded by the chip

circuit. ( $Q_o$ ) . It is expressed as shown in equation (2) and equation (3)

$$\frac{1}{Q} = \frac{1}{Q_o} + \frac{1}{Q_{CHIP}} \quad (2)$$

$$Q = \frac{Q_o \times Q_{CHIP}}{Q_o + Q_{CHIP}} \quad (3)$$

$Q_o$  is the result of losses in the antenna coil and in the capacitor which maybe on chip.  $Q_o$  can be measured in a simple way using a spectrum or network analyzer. The condition must be met that during measurement  $V_{CHIP}$  is sufficiently low that the chip is not powered, thus  $V_{CHIP} < Diode\ Threshold\ Voltage$ .

The Power Dissipation of the chip in the tag ( $P_{CHIP}$ ) can be specified by the chip manufacturer. As  $P_{CHIP}$  depends on the drive level (the level of  $V_{CHIP\ rms}$ ) this drive level should be specified, for example the level at which the chip starts functioning  $V_{CHIP\ rms} = V_{MIN}$ . So the expression for ( $Q_{CHIP}$ )

$$Q_{CHIP} = 2\pi f C \frac{V_{CHIP}^2}{P_{CHIP}} \quad (4)$$

The performance of RFID tags also depends upon the tolerance on the resonance frequency of the tag ( $f_o$ ) which is caused by the deviation in the self inductance ( $L$ ) and the capacitance ( $C$ ). On assuming the specified tuned frequency equals the carrier frequency ( $f_o$ ) of the reader, the expression of frequency deviation  $\Delta f$  either can have a positive value or a negative value is given by:

$$\Delta f = \pm \frac{tolerance}{100} f_o \quad (5)$$

This frequency deviation and tolerance has a great influence on the minimum Magnetic field Strength ( $H_{\min}$ ) which can increase or decrease the inductive coupling i.e affects the detection of tags inside or outside the boundary of Near field. Due to this reason, it also affects the Far field. To optimize the minimum Magnetic field strength, the components used in RFID Tags such as the capacitor, self-inductance, and chips must have small tolerance values.

#### IV. CONCLUSION

The performance of RFID tags, for both near and far field is highly dependent on a number of factors that includes small tolerance values of tag components. This is important to help ensure the least amount of deviation of the magnetic field that indirectly affects the Q value, and in turn the tag performance. Other factors including the material and frequency combination as an important matching exercise for reliable tag detection. External environmental factors are equally important for read tag performance.

#### REFERENCES

- [1] "Microwave Engineering" by D M Pozar.
- [2] "RFID Handbook" by Klaus Finkenzeller
- [3] "RFID: The Whole Story " by Leah Genuario
- [4] "Antenna Theory: Analysis and Design" by C. A. Balanis
- [5] "Network Analysis and Synthesis" by Franklin Fa-Kun Kuo